

DHRUV RAAJEEV

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Education

Texas A&M University

Bachelors of Science: Computer Science

Expected Graduation: May 2028

Location: College Station, TX

- Freshman at Texas A&M University majoring in Computer Science
- Relevant Coursework: Data Structures and Algorithms, Discrete Structures for Computing, Program Design and Concepts, Computation (Python), Experimental Physics and Engineering Lab, Linear Algebra

Relevant Skills

Frameworks & Languages: Python, C++, FastAPI, SQL, HTML/CSS, TensorFlow, PyTorch, XGBoost, TypeScript, Tkinter

Tools & Platforms: GitHub, React, Jupyter, Google Colab, VS Code, Remix IDE, Cursor, Claude Code, Vercel, Node.js

Work Experience

University of North Texas: Smart Electronics Systems Laboratory (Remote)

Jul 2024 – Oct 2024

High-School Research Assistant

Denton, TX

- Researched integrating Single Sign-On (SSO) with Decentralized Identifiers (DIDs) to enhance authentication in Software Patch Delivery Systems, analyzing protocols like OAuth 2.0 and DID methods such as DID:ETHR.
- Utilized Ethereum IDE for contract-contract interactions with Solidity, developing a modular framework proposal.
- Developed and deployed a message board smart contract as a decentralized system in Remix's Ethereum IDE, implementing storage and retrieval functions with Solidity fundamentals.

Arytic

July 2025 – November 2025

Software Development Intern (Remote)

Austin, TX

- Implemented python-based parsing engines to parse and segment PDF resume content into structured JSON outputs, extracting contact fields and matching against a 220+-term skills taxonomy for system interoperability.
- Engineered a semantic-search pipeline in Pinecone, upserting 384-dim BERT embeddings (all-MiniLM-L6-v2, Hugging Face Transformers) for cosine-similarity resume retrieval against job-description queries.
- Prototyped keyword- vs. embedding-based retrieval variants to evaluate candidate-to-role matching quality.

Maroon Fund

Jan 2026 – Present

Quantitative Developer

College Station, TX

- Backtested combined buy/sell signal pipeline (NVDA, AMZN, META) — walk-forward training simulation, trade log sanity checks, standardized local CSV inputs feeding 20+ performance metrics (Sharpe/sortino, drawdown, win rate, profit factor).
- Extended backtest stack with benchmark analytics versus SP 500 (alpha, beta, volatility, benchmark return) on 3-tier data fallback plus CLI improvements for repeatable runs at a student-managed quant fund with \$70K + AUM

Projects

Chest X-Ray Pneumonia Localization Architectures | [Source Code](#)

December 2023 - July 2024

- Built and compared CNN architectures (Keras ResNet50V2, InceptionV3) in Google Colab for pneumonia localization/detection on RSNA chest X-ray data, achieving 97% validation/test accuracy and IoU $\geq$ 0.700.
- Preprocessed 30,000 DICOM images with a custom data generator (flipping, interpolation, RGB channel stacking, resizing) to improve model robustness across architectures.

Advanced Log Analysis with SQLite Database Integration | [Source Code](#)

March 2025 - June 2025

- Developed a full-stack Python desktop application (Tkinter UI, SQLite storage) with a comprehensive engine to load, analyze, and visualize log data (2,000+ entries) via real-time regex-based pattern detection for errors, warnings, and other metrics.
- Implemented robust database management features including data visualization with sortable treeview components, text highlighting capability, and JSON session export functionality for potential external system integration.

Brittle: Interactive Stock Portfolio Fragility Playground | [Source Code](#)

May 2026 - July 2026

- Deployed a layered full-stack tool collapsing any ticker/weight set into a 0–100 Fragility Score, engineered from correlation, HHI concentration, PCA hidden-factor share (NumPy SVD/PC1), drawdown, volatility, and age-weighted VaR/CVaR across 3 weighted risk branches.
- Modeled single-point-of-failure risk via a 30% PC1-exposure shock with correlation-aware contagion, quantifying loss amplification vs. weight alone; hardened with 37 network-free pytest cases and per-ticker price caching.